

Enterprise Network Troubleshooting Lab

Cisco Packet Tracer Project Report

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1. Project Overview

This project is a Cisco Packet Tracer network troubleshooting lab that represents a small multi-site enterprise network with a head office and a branch site. The network uses VLAN segmentation, router-on-a-stick inter-VLAN routing, DHCP, OSPF dynamic routing, ACL-based traffic control, and SSH-based management access.

The main purpose of the project is not only to build a working network, but also to intentionally introduce realistic network faults, diagnose them using Cisco IOS commands, apply the correct fixes, and verify that services are restored.

Project objectives:

- Configure a small enterprise-style network using Cisco Packet Tracer.
- Implement VLANs, trunks, router subinterfaces, DHCP, OSPF, ACLs, and SSH.
- Create seven realistic troubleshooting tickets.
- Diagnose each issue using appropriate Cisco verification commands.
- Document the symptoms, root cause, fix, and verification for each fault.
- Prepare the project as a network engineering portfolio item.

2. Network Design Summary

2.1 Topology Description

- R1 acts as the head office router and provides inter-VLAN routing for HQ VLANs.
- R2 acts as the branch router and provides inter-VLAN routing for branch VLANs.
- S1 is the head office access switch, connected to R1 using an 802.1Q trunk.
- S2 is the branch access switch, connected to R2 using an 802.1Q trunk.
- R1 and R2 are connected using a point-to-point router link on the 10.0.0.0/30 network.
- OSPF area 0 is used to exchange routes between the head office and branch networks.

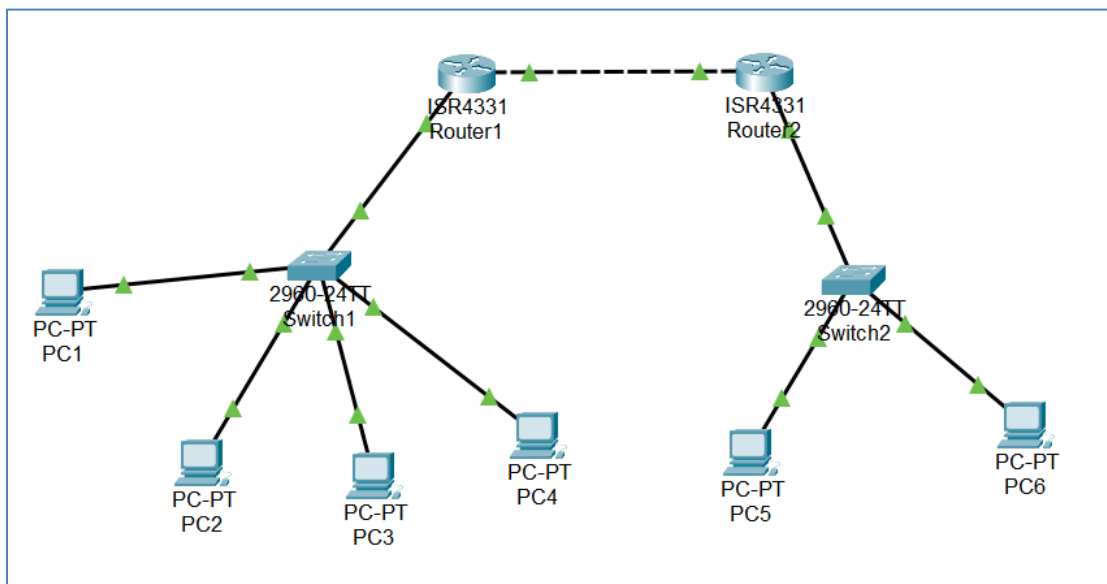


Fig 1: Full Topology

2.2 VLAN and IP Addressing Plan

VLAN	Name	HQ Subnet	Branch Subnet	Purpose
10	HR	192.168.10.0/24	192.168.110.0/24	HR users
20	IT	192.168.20.0/24	192.168.120.0/24	IT users
30	Guest	192.168.30.0/24	192.168.130.0/24	Guest users
99	Management	192.168.99.0/24	N/A	Device management
N/A	Router Link	10.0.0.0/30	10.0.0.0/30	R1 to R2 OSPF link

3. Baseline Working Network

Before introducing faults, the network was tested to confirm the baseline configuration worked correctly. This step is important because it proves that any later failure is caused by the intentionally introduced fault rather than an incomplete original setup.

Feature	Expected Result	Status
DHCP	PCs receive IP addresses from the correct VLAN pools	Working before faults
VLANs	Access ports are assigned to the correct VLANs	Working before faults
Trunks	Router-to-switch links carry required VLANs	Working before faults
Inter-VLAN Routing	Devices can reach allowed local VLAN gateways	Working before faults
OSPF	R1 and R2 exchange remote routes	Working before faults
ACL	HR VLAN 10 is blocked from IT VLAN 20	Working before faults
SSH	Management VLAN device can access R1 using SSH	Working before faults

4. Troubleshooting Methodology

Each troubleshooting ticket followed a consistent process. This approach reflects a practical helpdesk or junior network engineer workflow: confirm the issue, gather evidence, isolate the fault domain, identify the root cause, apply the fix, and verify that the service works again.

1. Identify the affected device, VLAN, or service.
2. Reproduce the fault using ping, DHCP renewal, or SSH testing.
3. Use show commands to inspect VLANs, trunks, routes, DHCP bindings, ACLs, or interfaces.
4. Compare the faulty configuration against the expected design.
5. Apply the minimum required fix.
6. Verify the result with a successful test and screenshot evidence.

5. Troubleshooting Tickets

5.1 Ticket 1: DHCP Failure

Problem

PC1 in VLAN 10 could not receive an IP address from DHCP.

Symptoms

- PC1 failed to obtain a 192.168.10.x address automatically.
- R1 showed DHCP leases for other VLANs, but not for VLAN 10.
- PC1 could reach the gateway when a static IP was configured, proving the VLAN path was working.

Commands Used

```
show ip dhcp binding
show ip dhcp pool
show running-config | section dhcp
ping 192.168.10.1
```

Root Cause

The DHCP exclusion range was incorrectly configured earlier, causing VLAN 10 DHCP allocation to fail. After removing the incorrect exclusion, the VLAN 10 DHCP pool still needed to be recreated in Packet Tracer before PC1 could receive a lease.

Fix Applied

```
no ip dhcp pool VLAN10
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
```

Verification

PC1 successfully received a valid 192.168.10.x DHCP address with gateway 192.168.10.1.

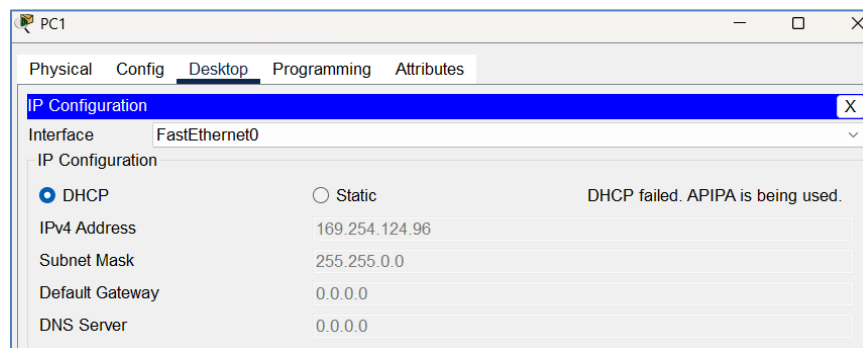


Fig 2.1: DHCP fail

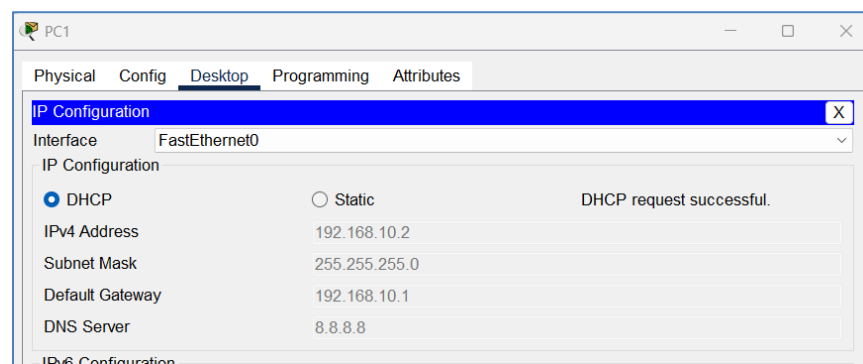


Fig 2.2: DHCP fixed

5.2 Ticket 2: Wrong VLAN Assignment

Problem

PC2 received an IP address from the wrong VLAN.

Symptoms

- PC2 was expected to be in VLAN 20.
- PC2 received an address from VLAN 30 instead.
- This caused PC2 to behave like a Guest VLAN device instead of an IT VLAN device.

Commands Used

```
show vlan brief
show interfaces fa0/3 switchport
```

Root Cause

The switch access port connected to PC2 was assigned to VLAN 30 instead of VLAN 20.

Fix Applied

```
interface fa0/3
  switchport mode access
  switchport access vlan 20
```

Verification

PC2 successfully received a 192.168.20.x DHCP address after the port was returned to VLAN 20.

☒ DHCP	☐ Static
IPv4 Address	192.168.30.3
Subnet Mask	255.255.255.0
Default Gateway	192.168.30.1
DNS Server	8.8.8.8

Fig 3.1: VLAN address fail

☒ DHCP	☐ Static
IPv4 Address	192.168.20.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	8.8.8.8

Fig 3.2: VLAN address fixed

5.3 Ticket 3: Trunk Allowed VLAN Issue

Problem

PC3 in Guest VLAN 30 could not reach its default gateway.

Symptoms

- PC3 had a VLAN 30 address.
- Ping to 192.168.30.1 failed.
- Other VLANs continued to work, which narrowed the issue to VLAN 30 forwarding.

Commands Used

```
show interfaces trunk
show vlan brief
```

Root Cause

VLAN 30 was configured on the switch and PC3 was correctly assigned to it, but VLAN 30 was missing from the trunk allowed VLAN list on S1.

Fix Applied

```
interface fa0/1
 switchport trunk allowed vlan 10,20,30,99
```

Verification

PC3 successfully pinged 192.168.30.1 after VLAN 30 was restored to the trunk allowed list.

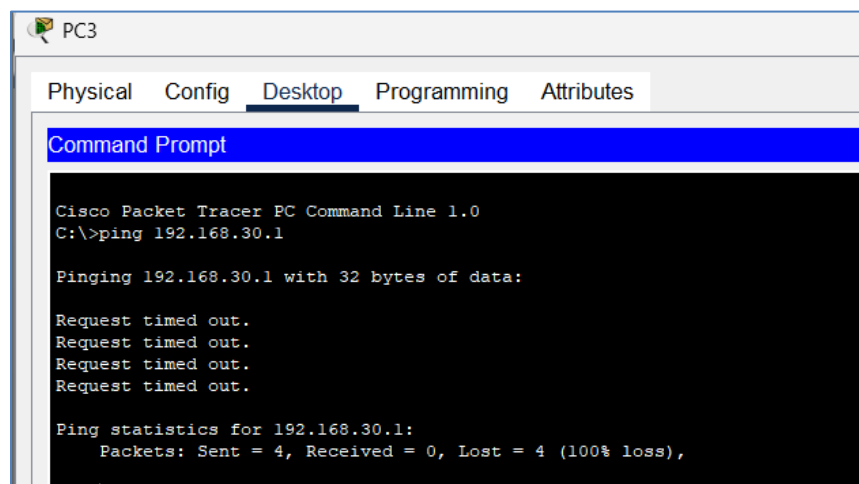


Fig 4.1: Trunk fail

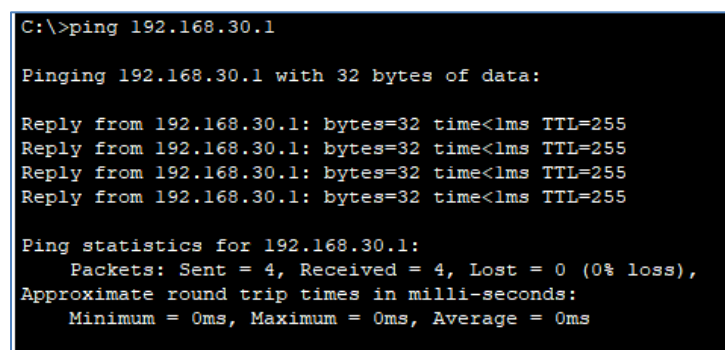


Fig 4.2: Trunk fixed

5.4 Ticket 4: OSPF Routing Failure

Problem

Head office and branch networks could not communicate because OSPF routing failed.

Symptoms

- PC1 could not reach the branch gateway.
- Remote branch routes were missing from the routing table.
- The OSPF neighbour relationship was missing or unstable.

Commands Used

```
show ip ospf neighbor
show ip route
show ip protocols
```

Root Cause

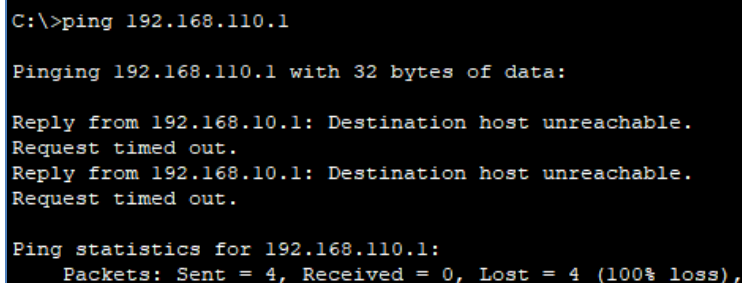
The R2 point-to-point link was placed in OSPF area 1 instead of area 0. The shared link had to be in the same OSPF area on both routers.

Fix Applied

```
router ospf 1
no network 10.0.0.0 0.0.0.3 area 1
network 10.0.0.0 0.0.0.3 area 0
```

Verification

R1 and R2 formed an OSPF neighbour relationship and remote routes appeared in the routing table.



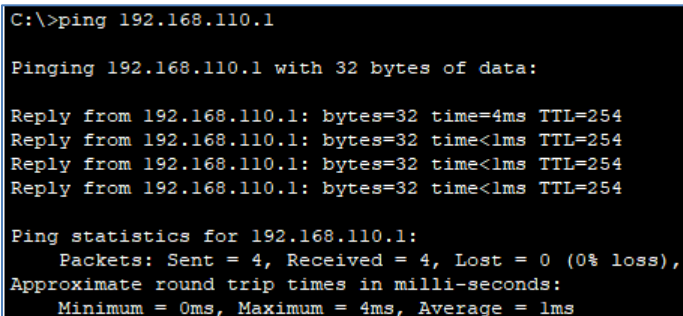
```
C:\>ping 192.168.110.1

Pinging 192.168.110.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Request timed out.
Reply from 192.168.10.1: Destination host unreachable.
Request timed out.

Ping statistics for 192.168.110.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig 5.1: OSPF fail



```
C:\>ping 192.168.110.1

Pinging 192.168.110.1 with 32 bytes of data:

Reply from 192.168.110.1: bytes=32 time=4ms TTL=254
Reply from 192.168.110.1: bytes=32 time<1ms TTL=254
Reply from 192.168.110.1: bytes=32 time<1ms TTL=254
Reply from 192.168.110.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.110.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

Fig 5.2: OSPF fixed

5.5 Ticket 5: ACL Misconfiguration

Problem

HR traffic was blocked from the wrong destination network.

Symptoms

- HR VLAN 10 was blocked from Guest VLAN 30.
- HR VLAN 10 was able to reach IT VLAN 20, which should have been restricted.
- The observed security behaviour did not match the intended policy.

Commands Used

```
show access-lists
show running-config | section access-list
show running-config interface g0/0/0.10
```

Root Cause

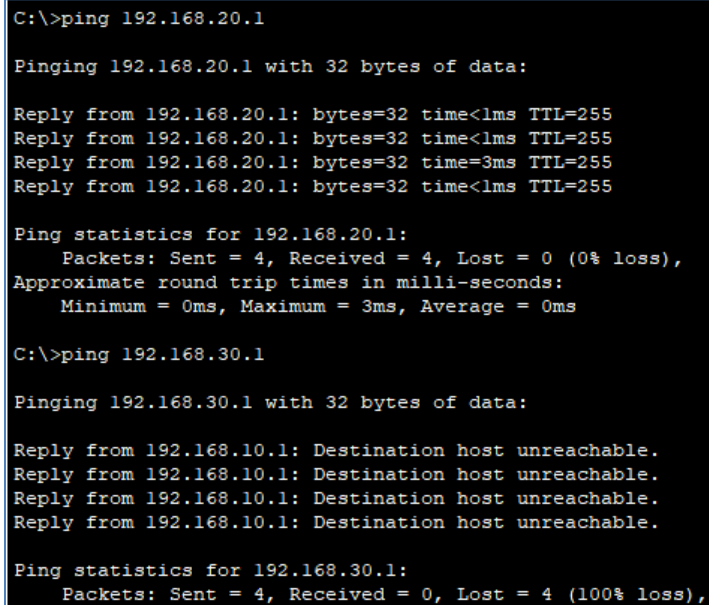
ACL 100 had the wrong destination network. It blocked HR to Guest instead of blocking HR to IT.

Fix Applied

```
no access-list 100
access-list 100 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 100 permit ip any any
```

Verification

HR to IT traffic was blocked, while HR to Guest traffic was allowed.



```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=3ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig 6.1: ACL fail

```

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Fig 6.2: ACL fixed

5.6 Ticket 6: SSH Management Access Failure

Problem

PC4 in the management VLAN could not SSH into R1.

Symptoms

- PC4 had the correct management VLAN IP address.
- SSH connection to R1 failed.
- The issue affected management access rather than normal data-plane connectivity.

Commands Used

```

show access-lists
show running-config | section line vty
show running-config | include access-list 10

```

Root Cause

The VTY access-class ACL allowed the wrong subnet. It allowed 192.168.98.0/24 instead of the management subnet 192.168.99.0/24.

Fix Applied

```

no access-list 10
access-list 10 permit 192.168.99.0 0.0.0.255

```

Verification

PC4 successfully accessed R1 using SSH after the ACL was corrected.

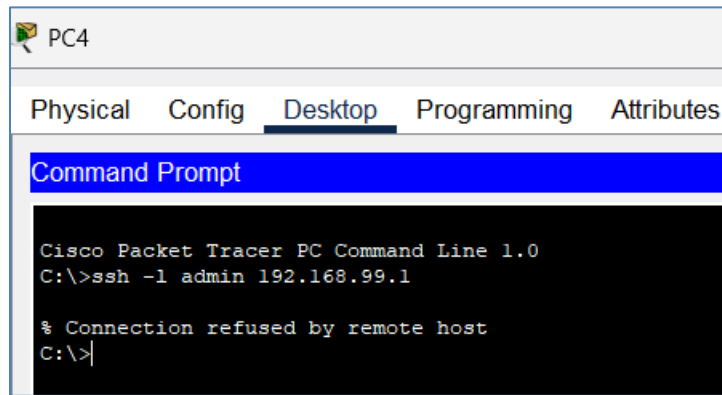


Fig 7.1: SSH fail

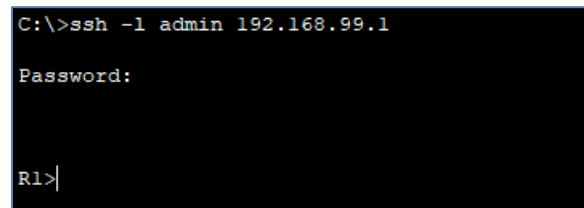


Fig 7.2: SSH fixed

5.7 Ticket 7: Branch Connectivity Issue

Problem

PC5 in the branch HR VLAN could not reach its gateway or HQ networks.

Symptoms

- PC5 was correctly connected to S2.
- PC5 could not ping 192.168.110.1.
- Branch connectivity to HQ failed.

Commands Used

```
show vlan brief
show interfaces trunk
show running-config interface g0/0/0.10
show ip interface brief
```

Root Cause

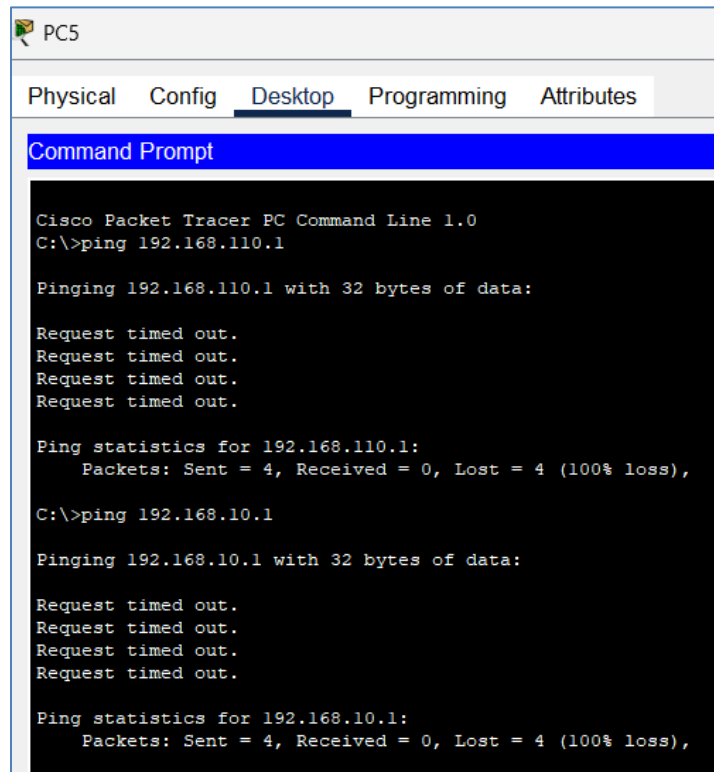
R2's VLAN 10 subinterface was configured with the wrong 802.1Q tag. PC5 was in VLAN 10, but R2 was expecting VLAN 11 traffic.

Fix Applied

```
interface g0/0/0.10
 encapsulation dot1Q 10
```

Verification

PC5 successfully pinged its default gateway and HQ networks after the VLAN tag was corrected.



PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.110.1

Pinging 192.168.110.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.110.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

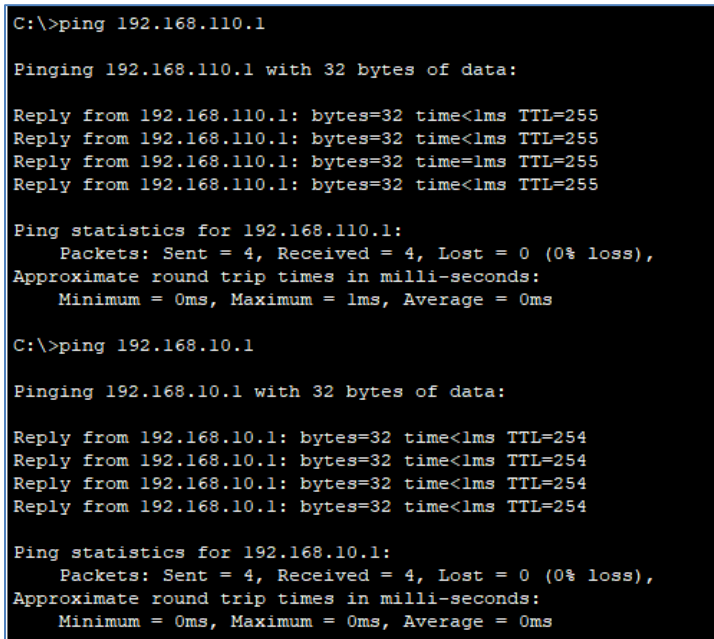
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Fig 8.1: Branch connection fail



```
C:\>ping 192.168.110.1

Pinging 192.168.110.1 with 32 bytes of data:

Reply from 192.168.110.1: bytes=32 time<1ms TTL=255
Reply from 192.168.110.1: bytes=32 time<1ms TTL=255
Reply from 192.168.110.1: bytes=32 time=1ms TTL=255
Reply from 192.168.110.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.110.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=254
Reply from 192.168.10.1: bytes=32 time<1ms TTL=254
Reply from 192.168.10.1: bytes=32 time<1ms TTL=254
Reply from 192.168.10.1: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig 8.2: Branch connection fixed

6. Final Verification

After all seven faults were resolved, the network was tested again to confirm that all expected services and connectivity paths were restored.

Test Area	Verification Test	Expected Result
DHCP	Renew DHCP on all PCs	Each PC receives an address from the correct VLAN subnet
VLANs	show vlan brief	Access ports appear in the correct VLANs
Trunking	show interfaces trunk	Required VLANs are allowed on trunk links
OSPF	show ip ospf neighbor	R1 and R2 show FULL neighbour state
Routing	show ip route	Remote networks appear as OSPF routes
ACL	Ping from HR to IT and HR to Guest	HR to IT blocked; HR to Guest allowed
SSH	SSH from PC4 to R1	Login succeeds from management VLAN

7. Lessons Learned

- A working baseline is important before troubleshooting because it provides a known-good reference point.
- DHCP issues can be caused by address pool problems, exclusions, bindings, VLAN paths, or switchport configuration.
- Wrong VLAN assignments can look like DHCP problems because the client may receive an address from the wrong subnet.
- Trunk allowed VLAN lists must match the VLANs required by the network design.
- OSPF neighbours require matching network statements, areas, and active interfaces on the shared link.
- ACLs should be checked carefully because a small destination network mistake can change the security policy.
- Management access should be separated from user traffic and restricted to trusted subnets.
- Branch connectivity problems can occur even when switching is correct if router subinterface tags are wrong.